# HTML5 - Canvas

**Benefits of HTML5**

1. Cleaner Code Structure
2. Cross-Browser Compatibility
3. Audio and Video Tags
4. Offline Browsing
5. No More Cookies
6. Canvas for Animation and Game Development

**HTML5 – Canvas** allows for dynamic, scriptable rendering of 2D shapes and bitmap images.

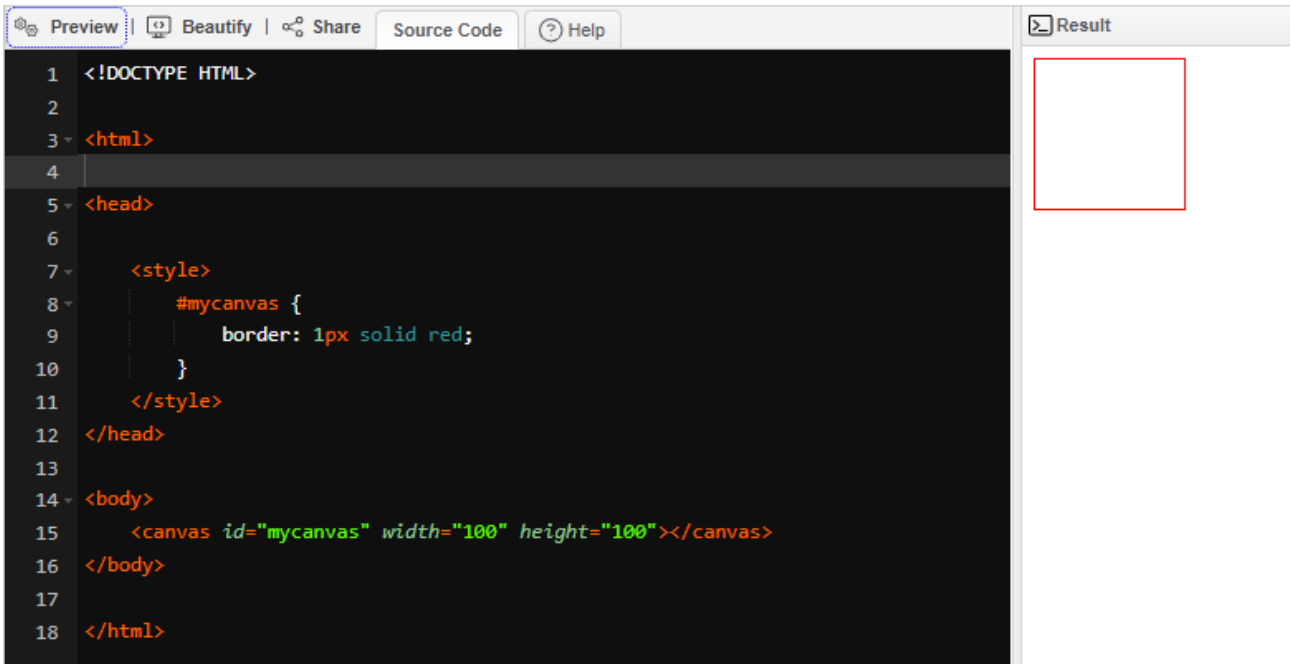
HTML canvas is a way to create illustrations and graphics using HTML and JavaScript. A canvas is created in HTML with the **<canvas>** element. Canvas illustrations can also move and can be made interactive, making them great for infographics, data visualizations, animations, and even simple games.

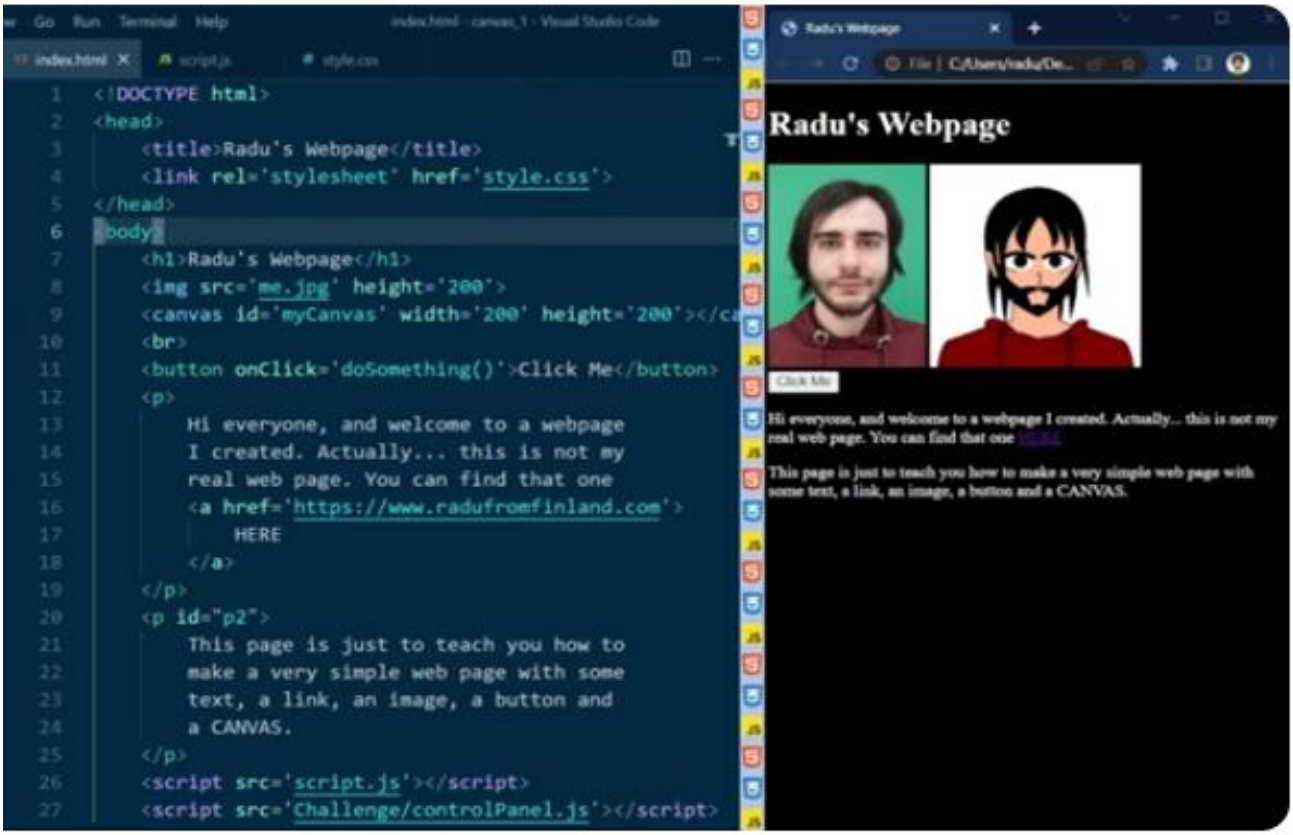
**Canvas element syntax**

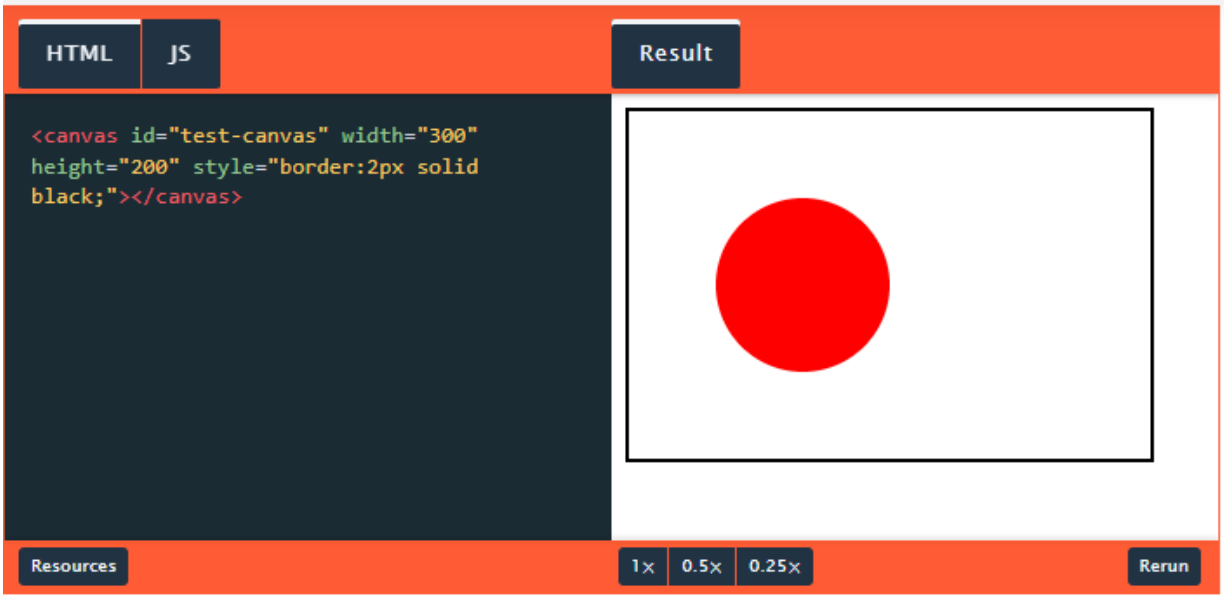
```
<canvas id = "mycanvas" width = "100" height = "100"></canvas>
```

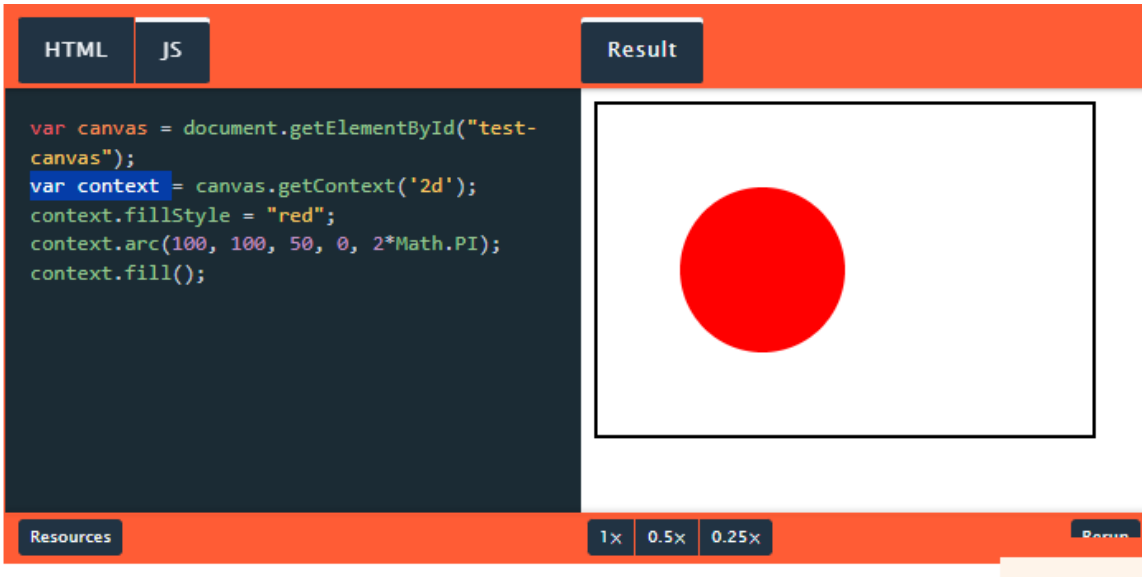
You can easily find that <canvas> element in the **DOM (**document object model**)**using getElementById() method as follows –

```
var canvas = document.getElementById("mycanvas");
```

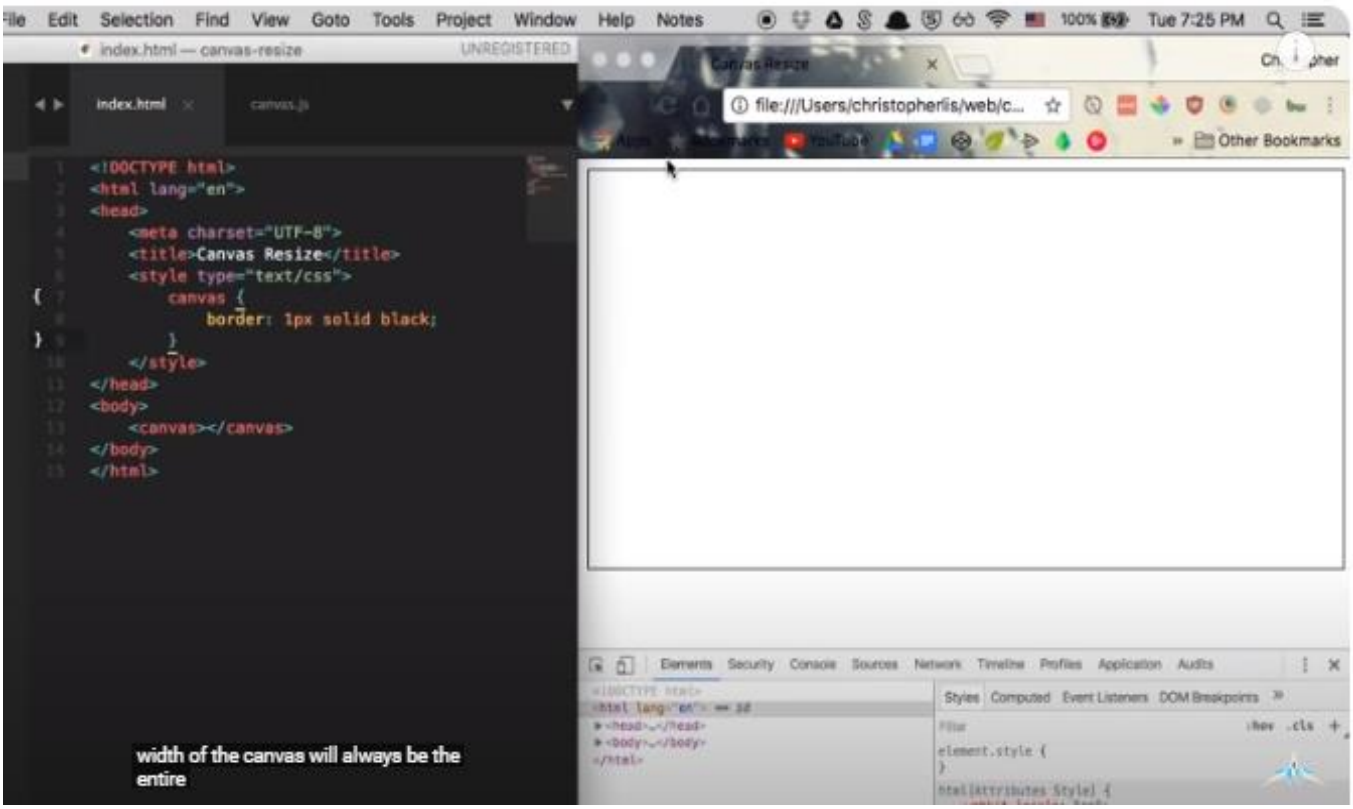


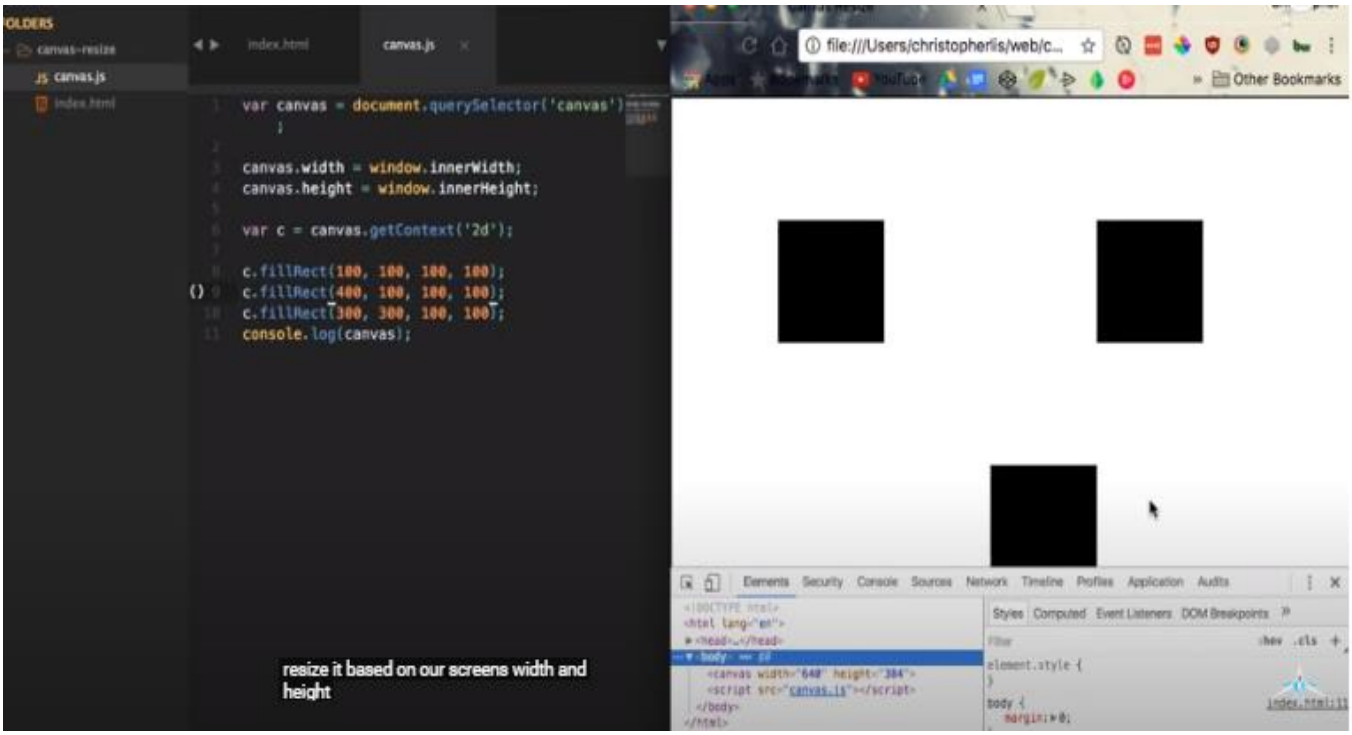

index.html
script.js
style.css
1 <!DOCTYPE html>
2 <head>
3 <title>Radu's Webpage</title>
4 <link rel='stylesheet' href='style.css'>
5 </head>
6 <body>
7 <h1>Radu's Webpage</h1>
8 <img src='me.jpg' height='200'>
9 <canvas id='myCanvas' width='200' height='200'></ca
10 <br>
11 <button onClick='doSomething()'>Click Me</button>
12 
13 Hi everyone, and welcome to a webpage
14 I created. Actually... this is not my
15 real web page. You can find that one
16 <a href='https://www.radufromfinland.com'>
17 HERE
18 </a>
19 
20 
21 This page is just to teach you how to
22 make a very simple web page with some
23 text, a link, an image, a button and
24 a CANVAS.
25 
26 <script src='script.js'></script>
27 <script src='Challenge/controlPanel.js'></script>
Radu's Webpage
Click Me
Hi everyone, and welcome to a webpage I created. Actually... this is not my real web page. You can find that one
This page is just to teach you how to make a very simple web page with some text, a link, an image, a button and a CANVAS.


HTML
JS
Result
<canvas id="test-canvas" width="300" height="200" style="border:2px solid black;"></canvas>
Resources
1×
0.5×
0.25×
Rerun




**Context variable**: A variable which can be set either at compile time or runtime. It can be changed and allows variables which would otherwise be hardcoded to be more dynamic.






**HTML5 Canvas Tutorial for Complete Beginners**
https://www.youtube.com/watch?v=EO6OkltgudE

**Drawing On HTML5 Canvas for Complete Beginners**
https://www.youtube.com/watch?v=83L6B13ixQ0

https://blog.hubspot.com/website/html-canvas

## **How to Create an HTML5 Game from Scratch** (02:16:10)?

https://www.youtube.com/watch?v=HpcWwopnnQE&t=4737s

00:40

Editor

04:55

Link Out to Javascript

21:57

Fill of the Rectangle

26:02

Adding in the Image

50:37

Event Listeners

01:02:10

Request Animation Frame

01:29:59

Random Calculations

01:37:27

Collision

01:55:07

Collision Detection

02:16:10

Source Code

